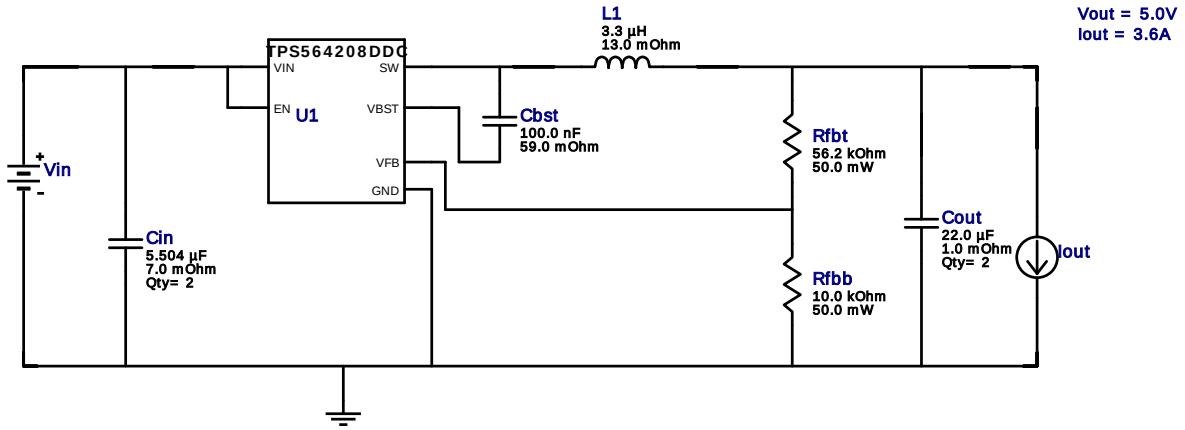
WEBENCH® Design Report

VinMin = 11.0V
VinMax = 16.0V
Vout = 5.0V
Iout = 3.6A

Device = TPS564208DDCR
Topology = Buck
Created = 2026-07-03 14:57:35.304
BOM Cost = $1.59
BOM Count = 9
Total Pd = 1.66W

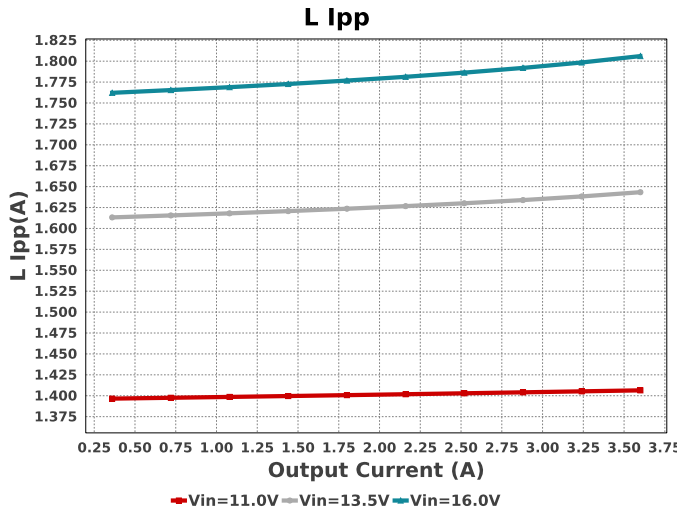
Design : 1 TPS564208DDCR
TPS564208DDCR 11V-16V to 5.00V @ 3.6A

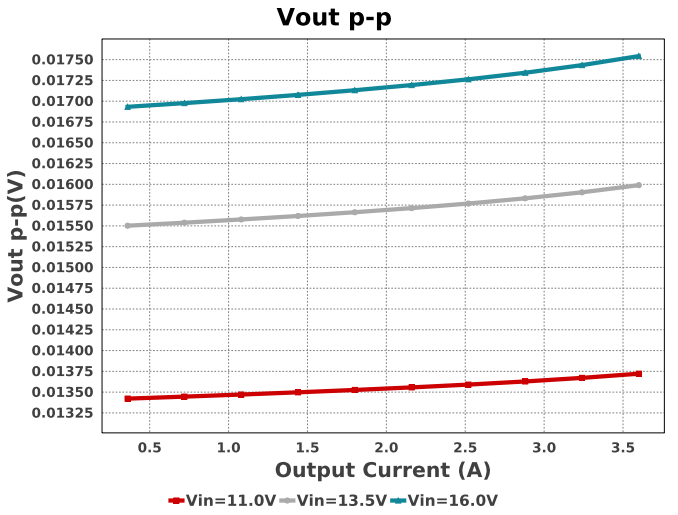


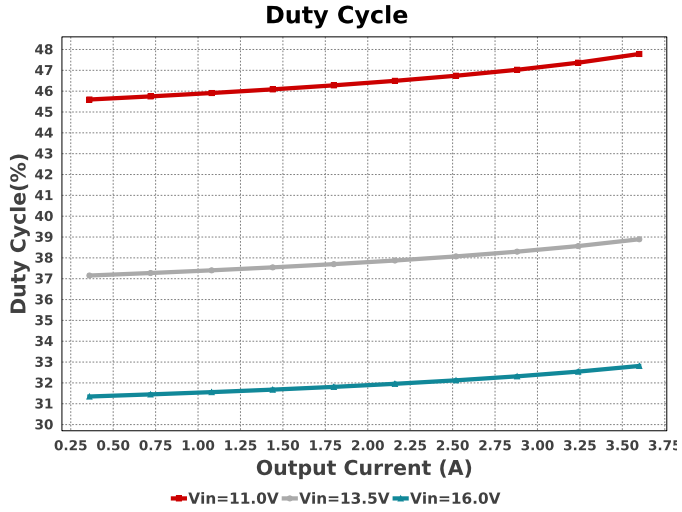
Electrical BOM

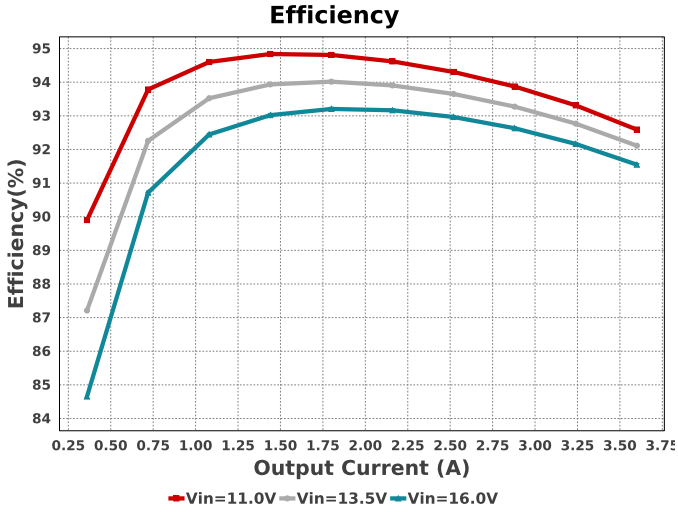
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbst	Kemet	C1206C104K5RACTU Series= X7R	Cap= 100.0 nF ESR= 59.0 mOhm VDC= 50.0 V IRMS= 1.2 A	1	$0.01	1206 11 mm²
Cin	MURATA	GRJ21BR61E106KE01K Series= X5R	Cap= 5.504 uF ESR= 7.0 mOhm VDC= 25.0 V IRMS= 2.14545 A	2	$0.07	0805 2 mm²
Cout	Taiyo Yuden	LMK212BJ226MG-T Series= X5R	Cap= 22.0 uF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 1.6 A	2	$0.09	0805 7 mm²
L1	VISHAY	IFDC5050HZER3R3N	L= 3.3 µH 13.0 mOhm	1	$0.75	IHLP-3232DZ 151 mm²
Rfbb	Yageo	RC0201FR-0710KL Series= ?	Res= 10.0 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	0201 2 mm²
Rfbt	Yageo	RC0201FR-0756K2L Series= ?	Res= 56.2 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	0201 2 mm²
U1	Texas Instruments	TPS564208DDCR	Switcher	1	$0.48	DDC0006A_N 10 mm²

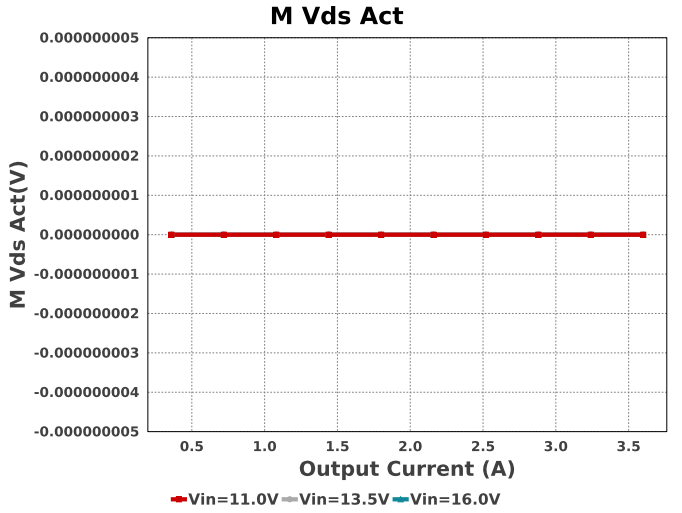
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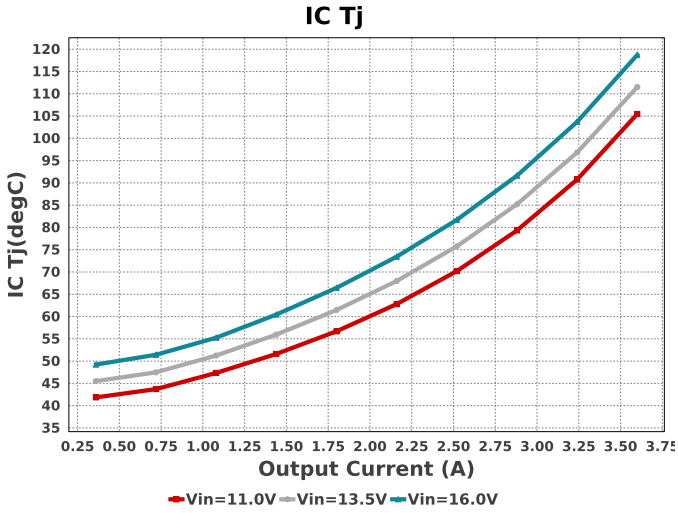






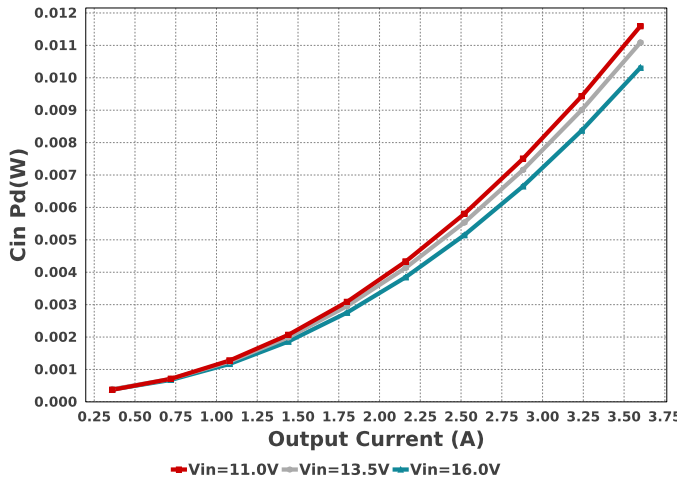




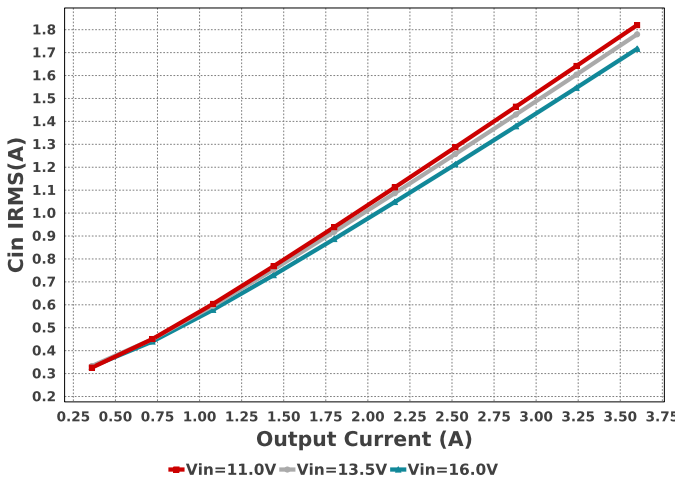


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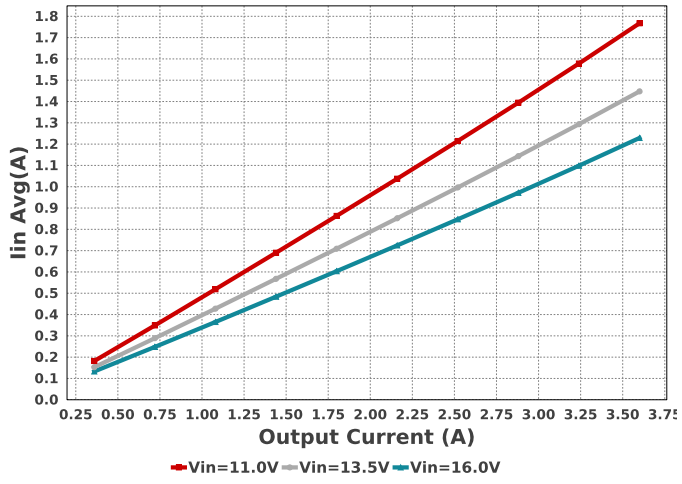
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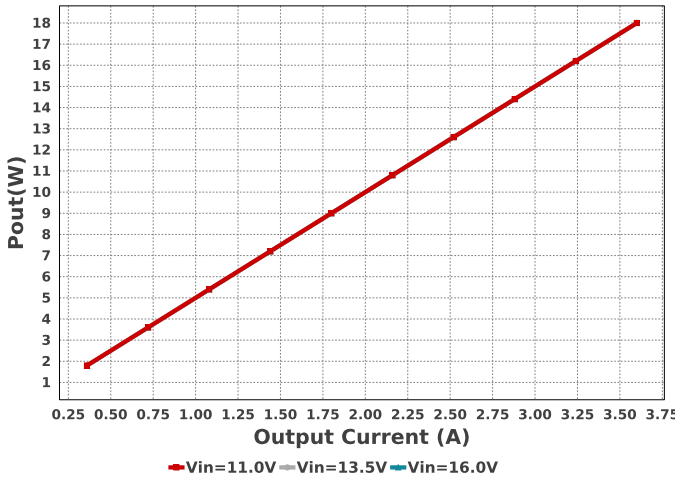
Cin IRMS



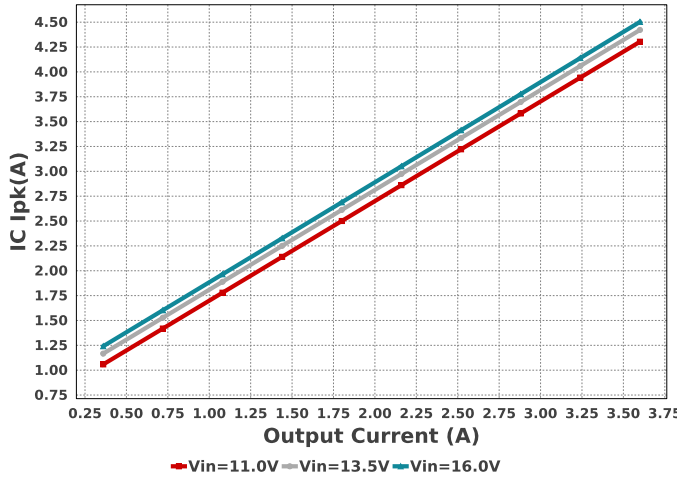
Iin Avg



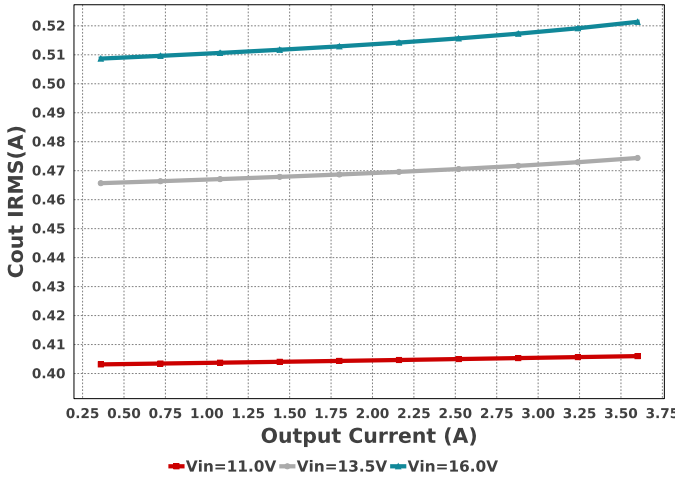
Pout



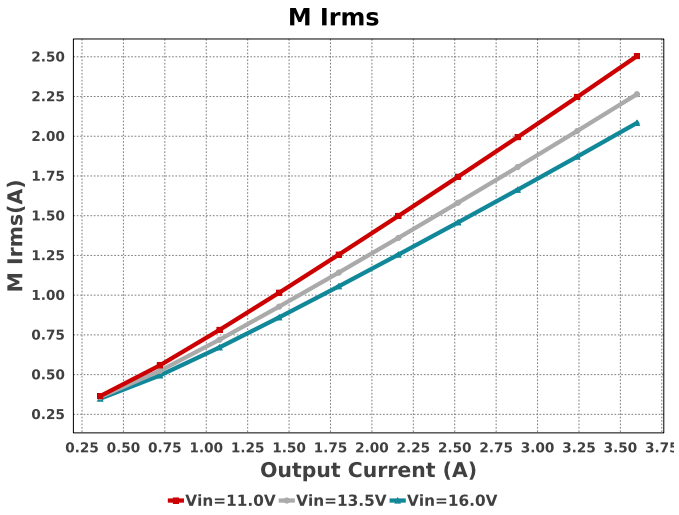
IC Ipk

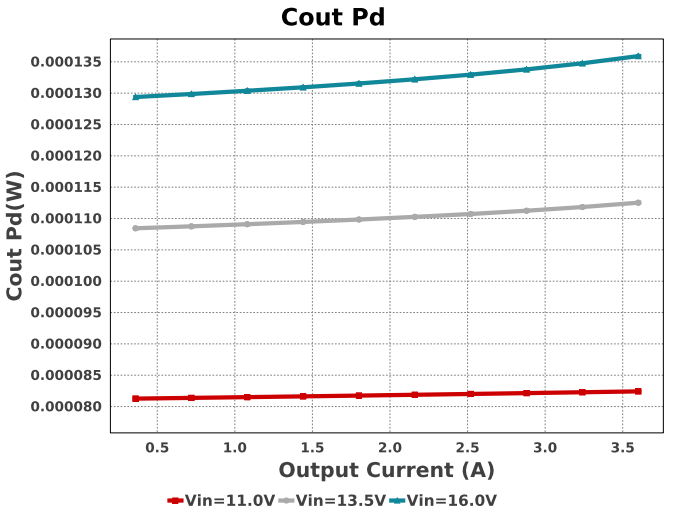


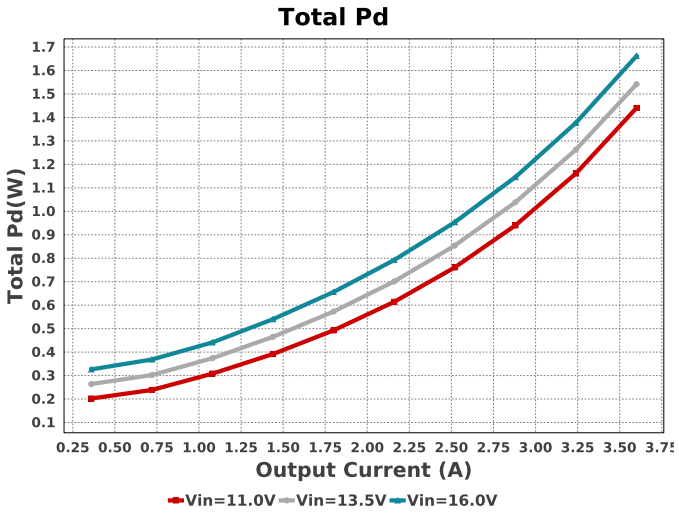
Cout IRMS

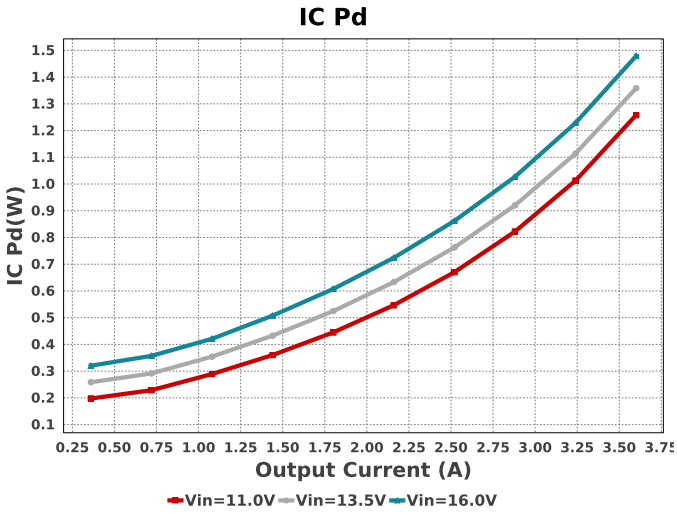


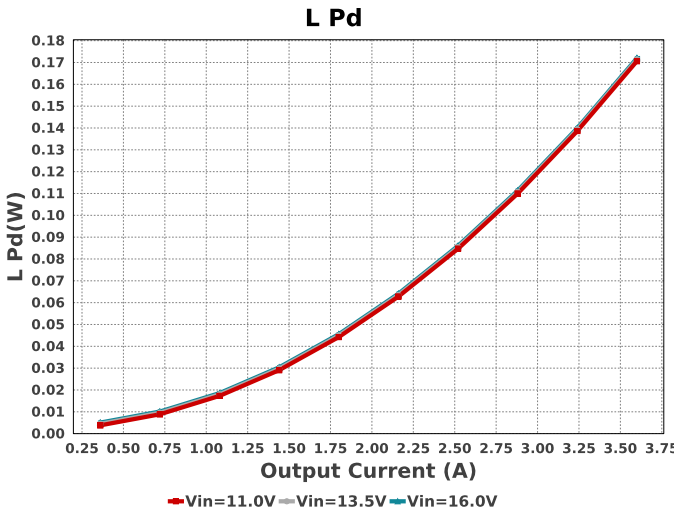
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Operating Values

#	Name	Value	Category	Description
1.	BOM Count	9		Total Design BOM count
2.	Total BOM	$1.588		Total BOM Cost
3.	Cin IRMS	1.716 A	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	10.312 mW	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	521.364 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	135.91 µW	Capacitor	Output capacitor power dissipation
7.	IC Ipk	4.503 A	IC	Peak switch current in IC
8.	IC Pd	1.479 W	IC	IC power dissipation
9.	IC Tj	118.738 degC	IC	IC junction temperature
10.	IC Tolerance	15.0 mV	IC	IC Feedback Tolerance
11.	ICThetaJA	60.0 degC/W	IC	IC junction-to-ambient thermal resistance

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#	Name	Value	Category	Description
12.	Iin Avg	1.229 A	IC	Average input current
13.	L Ipp	1.806 A	Inductor	Peak-to-peak inductor ripple current
14.	L Pd	172.01 mW	Inductor	Inductor power dissipation
15.	M1 Irms	2.084 A	Mosfet	Q Iavg
16.	M Vds Act	0.0 V	Mosfet	Voltage drop across the MosFET
17.	Cin Pd	10.312 mW	Power	Input capacitor power dissipation
18.	Cout Pd	135.91 µW	Power	Output capacitor power dissipation
19.	IC Pd	1.479 W	Power	IC power dissipation
20.	L Pd	172.01 mW	Power	Inductor power dissipation
21.	Total Pd	1.662 W	Power	Total Power Dissipation
22.	Duty Cycle	32.813 %	System Information	Duty cycle
23.	Efficiency	91.548 %	System Information	Steady state efficiency
24.	FootPrint	164.0 mm^2	System Information	Total Foot Print Area of BOM components
25.	Frequency	585.401 kHz	System Information	Switching frequency
26.	Iout	3.6 A	System Information	Iout operating point
27.	Mode	CCM	System Information	Conduction Mode
28.	Pout	18.0 W	System Information	Total output power
29.	Vin	16.0 V	System Information	Vin operating point
30.	Vout	5.0 V	System Information	Operational Output Voltage
31.	Vout Actual	5.031 V	System Information	Vout Actual calculated based on selected voltage divider resistors
32.	Vout Tolerance	3.723 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
33.	Vout p-p	17.543 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	3.6	Maximum Output Current
VinMax	16.0	Maximum input voltage
VinMin	11.0	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	TPS564208	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

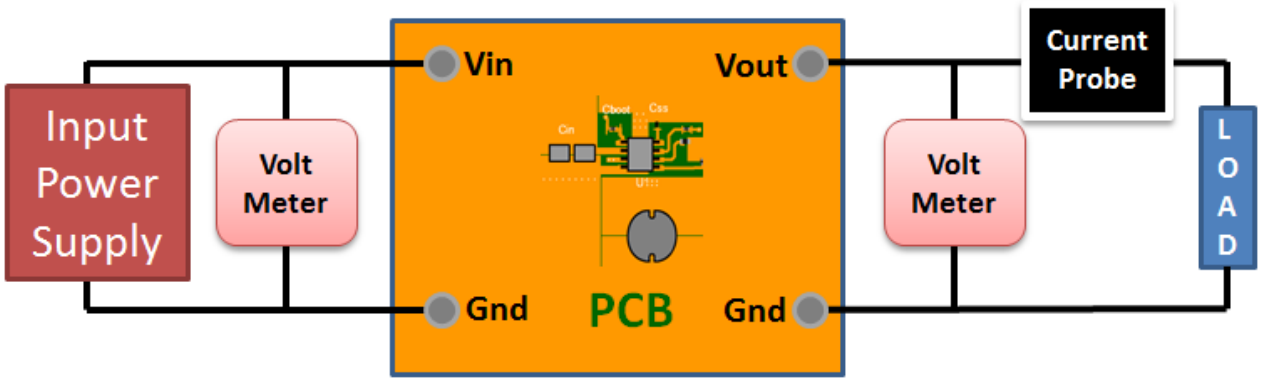
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 11.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



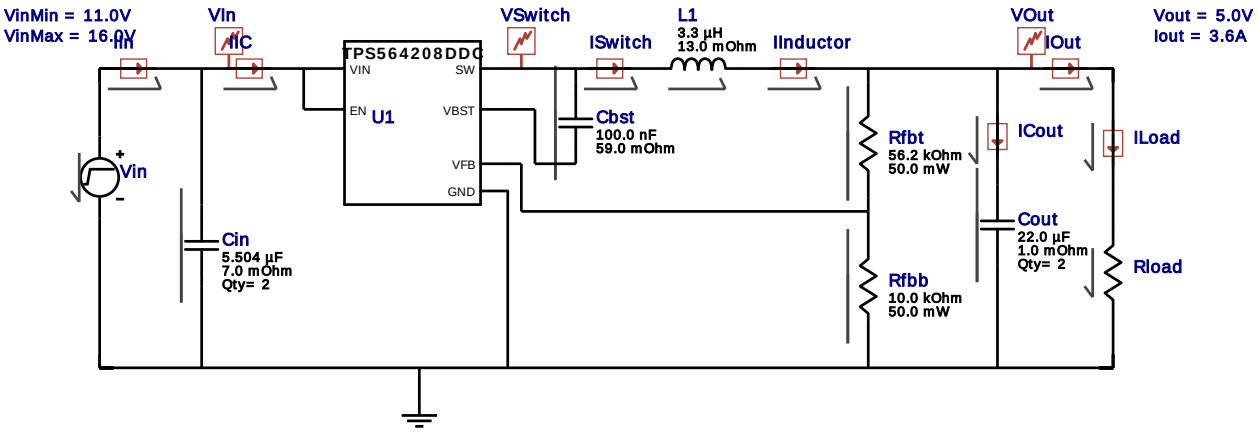
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WEBENCH® Electrical Simulation Report

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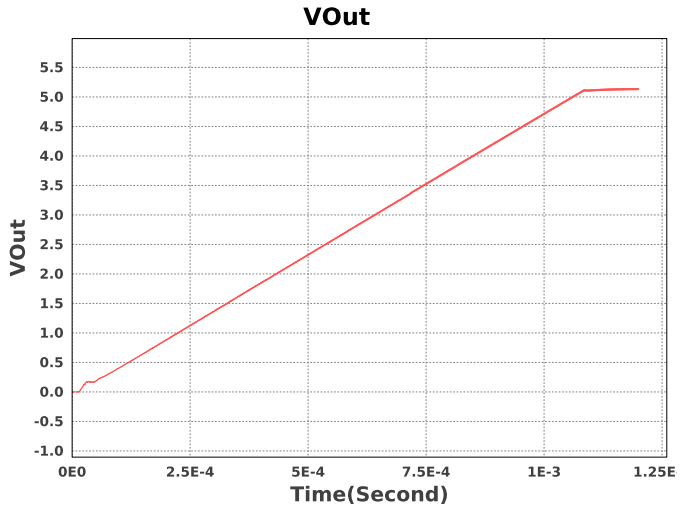
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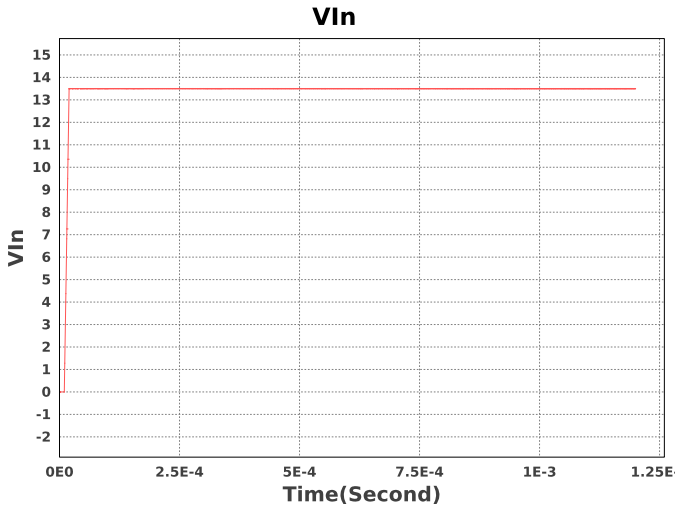
Simulation Type = Startup



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Rload	R	Load Resistance	1.3888888888888888 Ohm



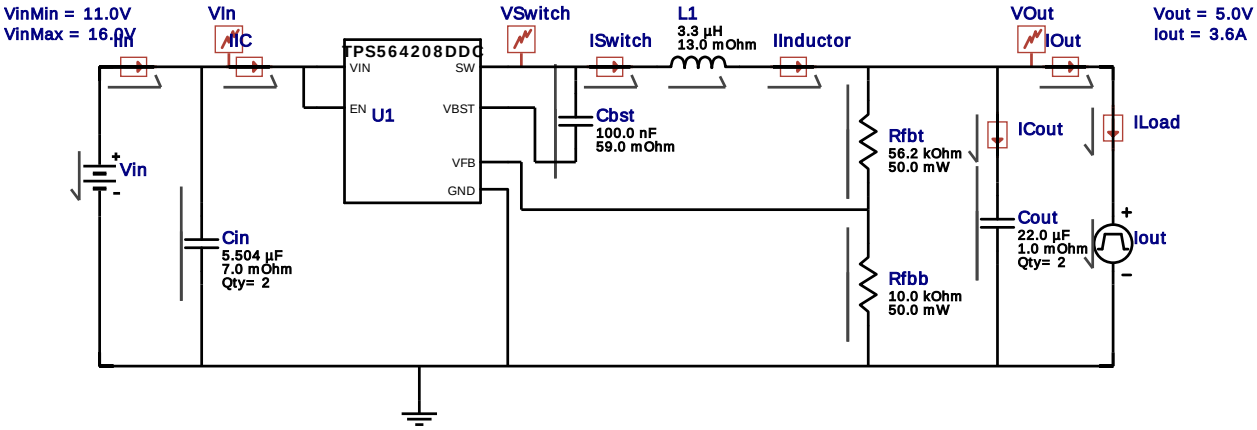


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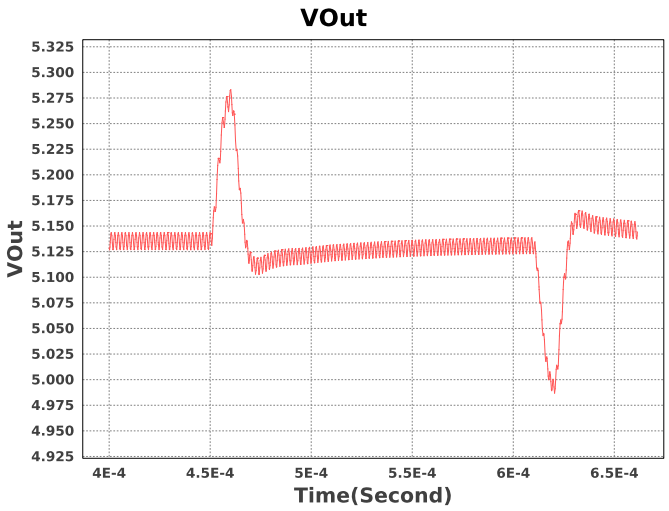
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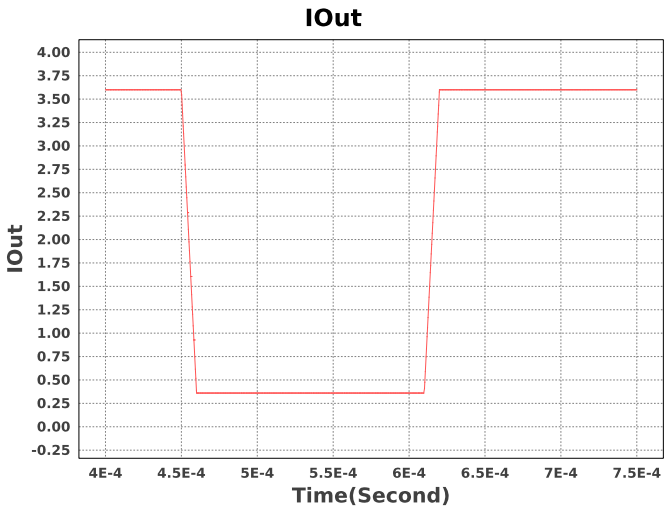
Simulation Type = Load Transient



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Cbst	IC	Initial Voltage	13.5 V
2.	Cout	IC	Initial Voltage	5.0 V
3.	L1	IC	Initial Current	0 V
4.	Iout	signal_type	Signal Type	PULSE
		I1	Initial Load Current	3.6 A
		I2	Minimum Load Current	0.36 A
		Td	Initial Time Delay	450u s
		Tf	Fall Time	10u s
		Tr	Rise Time	10u s
		Pw	Pulse Width	150u s



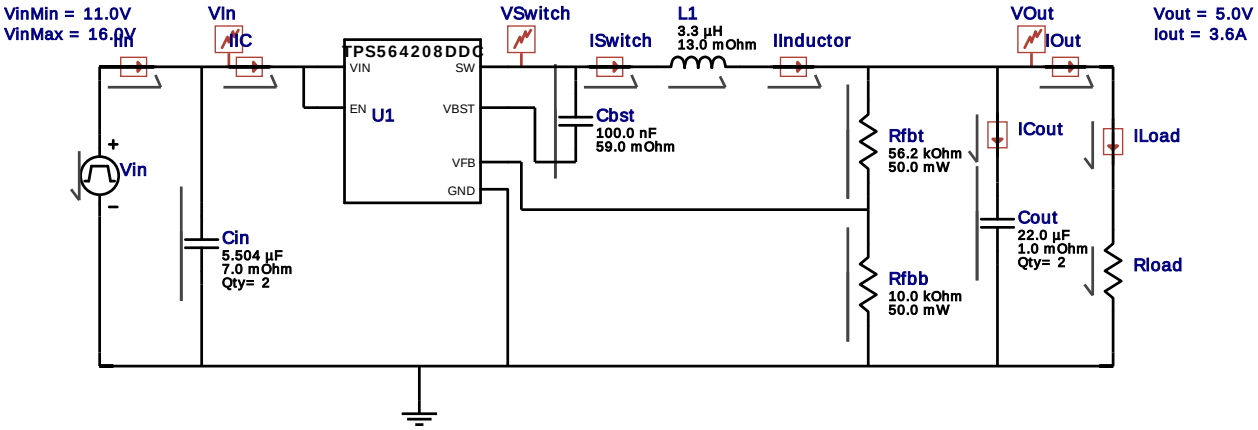


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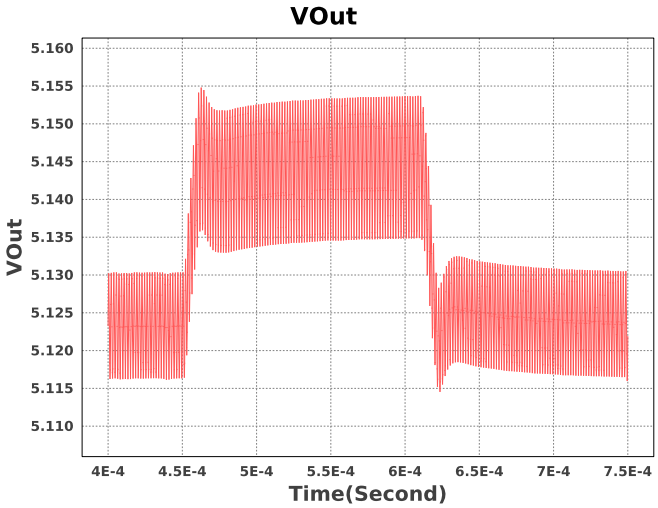
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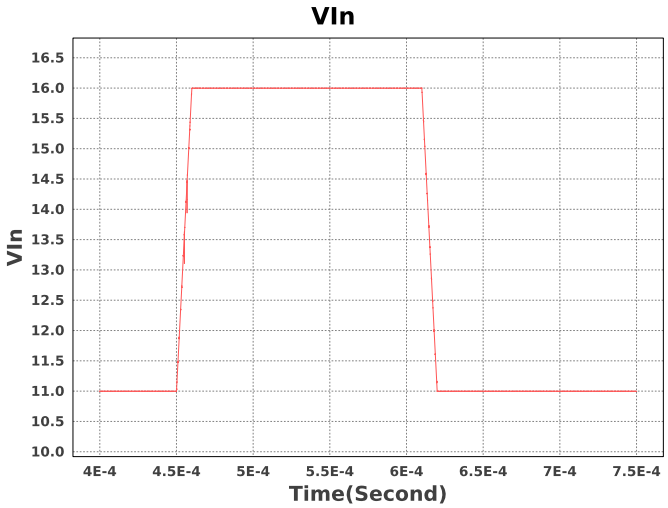
Simulation Type = Input Transient



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Cbst	IC	Initial Voltage	13.5 V
2.	Cout	IC	Initial Voltage	5.0 V
3.	L1	IC	Initial Current	0 V
4.	Rload	R	Load Resistance	1.3888888888888888 Ohm



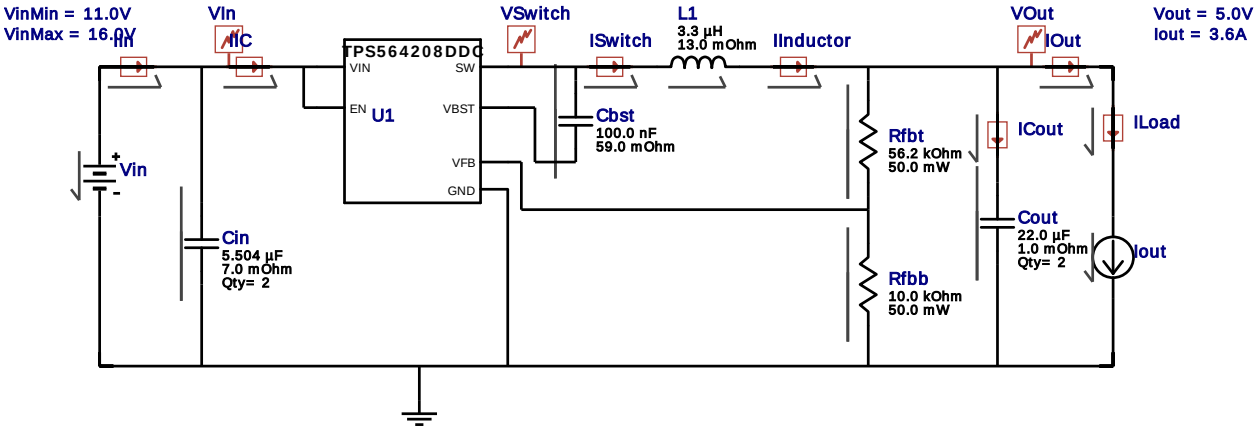


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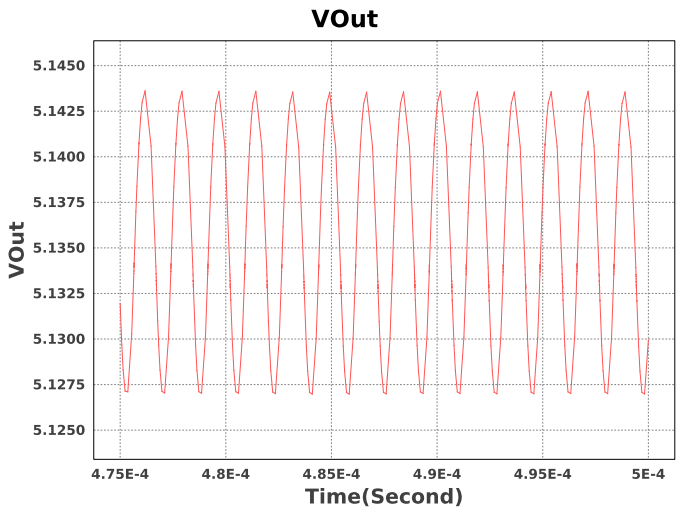
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Simulation Type = Steady State



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	IOut	IC	Initial Current	3.6 A
2.	Cbst	IC	Initial Voltage	13.5 V
3.	Cout	IC	Initial Voltage	5.0 V
4.	L1	IC	Initial Current	0 V
5.	Iout	I	Load Current	3.6 A



Design Assistance

1. Master key : E1F70865C65E4162399049D91329AE53[v1]
2. **TPS564208** Product Folder : http://www.ti.com/product/TPS564208 : contains the data sheet and other resources.

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